Swapnanil Bala

Boston, MA | +1 (617) 388-4577 | swapnanil.professionale@gmail.com
swapportfolio.vercel.app | github.com/SwapnanilBala | linkedin.com/in/swapnanil-bala-854b722a7

EDUCATION

Northeastern University, Khoury College of Computer Sciences Sep 2025 – May 2027 (expected)
M.S., Data Science | GPA 3.4/4.0
Coursework: Foundations of AI, Natural Language Processing, Machine Learning & Pattern Recognition, Algorithms, Essentials of Data Science, Data Mining Techniques

Maulana Abul Kalam Azad University of Technology (MAKAUT) 2019 – 2023
B.Tech., Electronics and Communications Engineering

Certifications: Google Advanced Data Analytics Certificate; Meta Database Engineer Professional Certificate

PROJECTS

Robust Health | app.robusthealth.in | *Next.js, TypeScript, Supabase, Claude API, Pinecone* May 2026 – Present
- Grounded AI plans in a RAG knowledge base (NIH supplement fact sheets, 48 Wikipedia muscle pages, ~870 exercises); the correct passage ranks first for 93% of 45 test questions and in the top 5 for 96%.
- Each AI plan adds up to 8 member-specific rules, each citing its source, at about $0.001 per plan; the AI revises the plan weekly from what the member actually logged.
- Kept a second, rule-based planner that makes no AI call: it leaves out exercises that clash with a member's injuries (knee, back, shoulder) and puts weak muscles first.
- CI on every push runs typecheck, lint, 1,312 tests, build and a performance budget; a smoke test confirms all 57 protected API routes reject signed-out requests.

Lagna Atelier | lagnaatelier.site | *TypeScript, Neon PostgreSQL, pgvector, Claude API* Jul 2026 – Present
- Grounded readings in three classical texts with hybrid retrieval (a chart-based filter, then pgvector similarity search) after early testers found them stale; their feedback then turned strongly positive.
- Moved generation from Claude Opus 5.5 to Haiku 5.5: a full PDF report went from ~186 s and $0.37 to 54-60 s and ~$0.006, with live progress shown while it writes.
- Added output guardrails in all 7 languages (script, banned-word and tone checks, one Opus retry on failure); rewording the prompt input cut English tone failures from 9 of 27 notes to 2 of 27.
- Screened typed questions for prompt injection and translated them to English before retrieval; cached responses by model input and ~2,100 system-prompt tokens per call.

Orthopedic Clinic Web App | drbalaortho.com | *Vercel, Neon PostgreSQL* Apr 2026 – Jun 2026
- Built a fill-in-the-blanks generator for print-ready prescriptions, used daily by a practicing orthopedic surgeon.
- Built booking and a patient portal in 3 languages: the doctor sets visit hours and approves patients, patients view and download past prescriptions, and the database keeps an audit trail.

BRFSS Health Dashboard | brfssdashanalysis.vercel.app | *Python, Plotly Dash, Pandas* Nov 2025 – Dec 2025
- Ported a course R/Shiny app to a live Plotly Dash dashboard over CDC BRFSS survey data (2011-present, 3.2M rows): pick any question and see its trend, state map and breakdowns by gender, age, race, education and income on one screen.
- Pre-computed every panel: the ~1 GB CDC file becomes a 21 MB Parquet file and 0.68 MB of JSON, so the app runs on Vercel as one serverless function; a verify step checks the fast pipeline against the original per-question code.

EXPERIENCE

Program Analyst Intern, Technology Nov 2024 – Jan 2025
P2G Mobility Tech Private Limited (Pointo) – EV-Financing Startup, Kolkata, India
- Cleaned operational data in Excel, built internal dashboards, and wrote basic SQL queries for recurring reports.
- Pitched pay-to-own EV financing to e-rickshaw operators on the ground as part of customer acquisition.

SKILLS

Languages: Python, TypeScript, JavaScript, SQL, R, C++ (familiar)
ML / AI: Claude API, RAG, pgvector, Pinecone, prompt caching, PyTorch, scikit-learn, Pandas, NumPy
Web / Backend: Next.js, FastAPI, OAuth
Databases: PostgreSQL (Neon, Supabase), MySQL
Tools: Git, CI/CD, Vercel